



Background

- Undergraduate Research Assistants (URAs) enter labs with varying levels of experience, requiring mentorship that is both structured and adaptable.
- Research skill development is incremental and benefits from clear guidance, as emphasized in the existing literature (Auchincloss et al., 2014; Willison & O'Regan, 2007).
- A lab-based promotion system supports URA growth by motivating students, fostering autonomy, and helping mentors set expectations across three developmental stages.

01

Research Assistant Trainee

- Begin with little to no research experience and focus on **learning lab responsibilities and research principles**.
- Build reliable **work ethics**, including punctuality, task completion, and professional communication.
- Learn core **research ethics** concepts: data integrity, informed consent, and confidentiality.
- Handle **basic problem-solving** tasks such as minor tech issues or logistical challenges.
- Develop **article reading skills** by learning the structure of journal articles and key research terminology.
- Strengthen collaboration through **peer teamwork**, mentor support, and use of shared tools like Google Drive or Slack.

02

Research Assistant

- Move beyond task-following to analyzing the **rationale behind study designs and methodologies**.
- Begin anticipating challenges and **proposing solutions** during experimental procedures.
- Critically evaluate literature**, interpret statistical results, and assess study strengths and limitations.
- Strengthen **quantitative skills** by understanding models, p-values, confidence intervals, and effect sizes.
- Improve **communication** by writing summaries, presenting in lab meetings, and contributing to team discussions.
- Develop deeper **critical thinking**, research expertise, and **collaborative confidence**.

Not Just for Course Credits

Developing Undergraduate Research Assistants Through a Lab Promotion System

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A framework to grow undergraduate researchers through intentional mentorship



03

Research Team Leader

- Lead by mentoring new URAs, organizing schedules, and **building a strong team environment**.
- Propose research ideas**, revise protocols, and streamline **lab operations** through innovation.
- Conduct in-depth literature reviews and **synthesize findings** across multiple studies.
- Deliver **advanced communication** via posters, proposals, and write effectively for varied audiences.
- Contribute to IRB processes, data management, and uphold **high ethical standards**.
- Demonstrate **independence** and **creativity** aligned with the highest stages of research skill development.

Implications and Conclusions

- A structured promotion system motivates URAs by recognizing their growth and giving them a **clear path to success**.
- Defined stages help URAs build **confidence**, develop **autonomy**, and master progressively complex research skills.
- Mentors benefit from having **clear expectations** and training strategies that align with each stage of student development.
- Leadership, critical thinking, and innovation are nurtured at higher levels, preparing URAs for **future academic and professional roles**.
- This approach cultivates a positive, productive lab environment and a stronger research culture overall.

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